

| Graph  | Query                   | Parameters      | Apache AGE   |                |                | Kuzu   |                |                | Neo4j   |                |                |
|--------|-------------------------|-----------------|--------------|----------------|----------------|--------|----------------|----------------|---------|----------------|----------------|
|        |                         |                 | B (ms)       | S <sub>D</sub> | S <sub>P</sub> | B (ms) | S <sub>D</sub> | S <sub>P</sub> | B (ms)  | S <sub>D</sub> | S <sub>P</sub> |
| F40    | <i>Q<sub>desc</sub></i> | Root            | 9.6          | 6.7x           | 6.9x           | 3.0    | 1.6x           | 1.6x           | 2.0     | 0.99x          | 0.88x          |
| F40    | <i>Q<sub>desc</sub></i> | Leaf parent H   | 8.8          | 6.3x           | 6.4x           | 2.7    | 1.9x           | 2.1x           | 1.3     | 0.86x          | 0.87x          |
| F40    | <i>Q<sub>desc</sub></i> | Leaf parent     | 8.9          | 6.4x           | 6.4x           | 2.7    | 1.4x           | 1.6x           | 1.3     | 0.86x          | 0.78x          |
| F1K    | <i>Q<sub>desc</sub></i> | Root            | 441.6        | 190x           | 190x           | 6.3    | 2.1x           | 2.5x           | 29.5    | 1.0x           | 0.71x          |
| F1K    | <i>Q<sub>desc</sub></i> | Leaf parent H   | 13.3         | 5.9x           | 5.9x           | 4.2    | 2.2x           | 1.9x           | 1.9     | 0.68x          | 0.85x          |
| F1K    | <i>Q<sub>desc</sub></i> | Leaf parent     | 13.0         | 6.7x           | 7.1x           | 3.6    | 1.7x           | 1.8x           | 1.8     | 1.0x           | 0.95x          |
| NT10   | <i>Q<sub>desc</sub></i> | Root            | 9.3          | 7.0x           | 6.9x           | 2.8    | 1.4x           | 1.6x           | 2.2     | 1.0x           | 0.96x          |
| NT10   | <i>Q<sub>desc</sub></i> | Leaf parent H/L | 9.3          | 7.0x           | 6.9x           | 2.6    | 1.5x           | 1.7x           | 1.4     | 0.83x          | 0.88x          |
| NT100  | <i>Q<sub>desc</sub></i> | Root            | 20.1         | 13x            | 13x            | 3.9    | 2.2x           | 2.2x           | 11.5    | 0.86x          | 0.84x          |
| NT100  | <i>Q<sub>desc</sub></i> | Leaf parent H   | 9.9          | 6.8x           | 7.0x           | 3.1    | 1.7x           | 1.9x           | 2.2     | 0.91x          | 0.94x          |
| NT100  | <i>Q<sub>desc</sub></i> | Leaf parent     | 8.8          | 6.1x           | 6.2x           | 2.9    | 1.9x           | 1.8x           | 1.5     | 0.95x          | 0.83x          |
| NT1K   | <i>Q<sub>desc</sub></i> | Root            | 1,114.5      | 400x           | 390x           | 10.1   | 5.2x           | 4.9x           | 57.7    | 1.3x           | 1.2x           |
| NT1K   | <i>Q<sub>desc</sub></i> | Leaf parent H   | 15.5         | 6.7x           | 6.8x           | 2.9    | 1.9x           | 1.7x           | 1.3     | 0.93x          | 0.90x          |
| NT1K   | <i>Q<sub>desc</sub></i> | Leaf parent     | 12.5         | 5.4x           | 5.5x           | 3.8    | 2.2x           | 2.1x           | 1.2     | 0.99x          | 0.90x          |
| NT10K  | <i>Q<sub>desc</sub></i> | Root            | 107,135.3    | 6.9Kx          | 6.9Kx          | 101.3  | 9.5x           | 11x            | 416.5   | 0.85x          | 0.78x          |
| NT10K  | <i>Q<sub>desc</sub></i> | Leaf parent H   | 269.0        | 23x            | 28x            | 2.7    | 0.56x          | 1.1x           | 1.4     | 1.0x           | 1.0x           |
| NT10K  | <i>Q<sub>desc</sub></i> | Leaf parent     | 245.4        | 21x            | 25x            | 3.0    | 0.51x          | 1.4x           | 1.2     | 0.67x          | 0.65x          |
| NT100K | <i>Q<sub>desc</sub></i> | Root            | 19,456,928.3 | 130Kx          | 140Kx          | 261.2  | 2.8x           | 3.4x           | 4,319.7 | 0.82x          | 0.81x          |
| NT100K | <i>Q<sub>desc</sub></i> | Leaf parent H   | 2,542.6      | 18x            | 20x            | 3.8    | 0.07x          | 0.17x          | 2.1     | 0.86x          | 0.65x          |
| NT100K | <i>Q<sub>desc</sub></i> | Leaf parent     | 469.9        | 3.4x           | 4.1x           | 2.5    | 0.05x          | 0.09x          | 1.4     | 0.90x          | 0.77x          |
| DT10   | <i>Q<sub>desc</sub></i> | Root            | 9.5          | 7.0x           | 6.7x           | 2.9    | 2.0x           | 1.7x           | 1.2     | 0.54x          | 0.53x          |
| DT10   | <i>Q<sub>desc</sub></i> | Leaf parent H   | 9.3          | 6.8x           | 6.8x           | 2.8    | 1.8x           | 1.8x           | 1.7     | 0.91x          | 0.89x          |
| DT10   | <i>Q<sub>desc</sub></i> | Leaf parent     | 9.0          | 6.6x           | 6.7x           | 2.8    | 1.8x           | 1.7x           | 1.3     | 0.97x          | 0.97x          |
| DT100  | <i>Q<sub>desc</sub></i> | Root            | 20.4         | 14x            | 14x            | 3.9    | 2.6x           | 2.2x           | 11.6    | 0.86x          | 0.84x          |
| DT100  | <i>Q<sub>desc</sub></i> | Leaf parent H   | 9.3          | 6.5x           | 6.6x           | 3.0    | 2.0x           | 1.8x           | 1.7     | 0.95x          | 0.95x          |
| DT100  | <i>Q<sub>desc</sub></i> | Leaf parent     | 9.3          | 6.5x           | 6.4x           | 2.8    | 1.8x           | 1.7x           | 1.5     | 0.94x          | 0.92x          |
| DT1K   | <i>Q<sub>desc</sub></i> | Root            | 1,107.9      | 390x           | 390x           | 11.1   | 6.6x           | 6.2x           | 39.8    | 0.84x          | 0.81x          |
| DT1K   | <i>Q<sub>desc</sub></i> | Leaf parent H   | 13.7         | 5.9x           | 6.2x           | 3.5    | 2.1x           | 2.2x           | 0.9     | 0.95x          | 0.99x          |
| DT1K   | <i>Q<sub>desc</sub></i> | Leaf parent     | 10.9         | 4.7x           | 4.9x           | 3.7    | 2.4x           | 2.4x           | 0.7     | 0.92x          | 0.69x          |
| DT10K  | <i>Q<sub>desc</sub></i> | Root            | 107,336.6    | 6.7Kx          | 7.0Kx          | 32.1   | 4.5x           | 3.6x           | 417.6   | 0.85x          | 0.82x          |
| DT10K  | <i>Q<sub>desc</sub></i> | Leaf parent H   | 246.8        | 21x            | 25x            | 2.6    | 0.56x          | 0.98x          | 1.2     | 0.89x          | 0.77x          |
| DT10K  | <i>Q<sub>desc</sub></i> | Leaf parent     | 223.2        | 19x            | 24x            | 2.6    | 0.49x          | 0.92x          | 1.4     | 0.93x          | 1.2x           |
| DT100K | <i>Q<sub>desc</sub></i> | Root            | >6 h         | >150Kx         | >150Kx         | 134.7  | 1.5x           | 1.7x           | 4,296.9 | 0.83x          | 0.80x          |
| DT100K | <i>Q<sub>desc</sub></i> | Leaf parent H   | 801.4        | 5.6x           | 6.6x           | 3.4    | 0.07x          | 0.13x          | 1.3     | 0.74x          | 0.76x          |
| DT100K | <i>Q<sub>desc</sub></i> | Leaf parent     | 473.4        | 3.4x           | 3.9x           | 2.3    | 0.04x          | 0.17x          | 1.6     | 0.99x          | 1.0x           |

| Graph  | Query        | Parameters    | Apache AGE   |       |       | Kuzu   |       |       | Neo4j   |       |       |
|--------|--------------|---------------|--------------|-------|-------|--------|-------|-------|---------|-------|-------|
|        |              |               | B (ms)       | SD    | SP    | B (ms) | SD    | SP    | B (ms)  | SD    | SP    |
| WT10   | <i>Qdesc</i> | Root          | 9.4          | 6.8x  | 6.6x  | 3.0    | 1.9x  | 2.0x  | 2.0     | 0.93x | 0.91x |
| WT10   | <i>Qdesc</i> | Leaf parent H | 9.3          | 6.6x  | 7.0x  | 3.1    | 1.9x  | 2.1x  | 1.7     | 0.89x | 0.85x |
| WT10   | <i>Qdesc</i> | Leaf parent   | 9.0          | 6.6x  | 6.6x  | 2.8    | 2.0x  | 1.9x  | 1.4     | 0.97x | 0.96x |
| WT100  | <i>Qdesc</i> | Root          | 26.0         | 12x   | 14x   | 3.5    | 2.3x  | 2.3x  | 11.6    | 0.88x | 0.86x |
| WT100  | <i>Qdesc</i> | Leaf parent H | 10.3         | 7.1x  | 7.1x  | 2.7    | 1.6x  | 1.9x  | 2.5     | 0.93x | 0.91x |
| WT100  | <i>Qdesc</i> | Leaf parent   | 11.7         | 6.7x  | 6.8x  | 3.0    | 2.0x  | 2.0x  | 1.7     | 0.82x | 0.95x |
| WT1K   | <i>Qdesc</i> | Root          | 1,152.5      | 440x  | 420x  | 8.7    | 5.3x  | 3.9x  | 58.8    | 1.3x  | 1.3x  |
| WT1K   | <i>Qdesc</i> | Leaf parent H | 20.2         | 8.8x  | 8.7x  | 3.3    | 1.7x  | 1.8x  | 1.6     | 0.77x | 0.60x |
| WT1K   | <i>Qdesc</i> | Leaf parent   | 21.9         | 9.7x  | 13x   | 3.5    | 2.2x  | 2.0x  | 2.2     | 0.93x | 0.88x |
| WT10K  | <i>Qdesc</i> | Root          | 112,250.6    | 13Kx  | 10Kx  | 24.9   | 2.4x  | 2.9x  | 407.5   | 0.84x | 0.81x |
| WT10K  | <i>Qdesc</i> | Leaf parent H | 332.9        | 29x   | 33x   | 2.4    | 0.47x | 0.81x | 1.7     | 0.78x | 0.61x |
| WT10K  | <i>Qdesc</i> | Leaf parent   | 114.7        | 20x   | 60x   | 3.0    | 0.65x | 1.0x  | 2.4     | 0.92x | 0.93x |
| WT100K | <i>Qdesc</i> | Root          | 23,750,824.0 | 310Kx | 320Kx | 122.5  | 1.3x  | 1.6x  | 4,313.0 | 0.84x | 0.82x |
| WT100K | <i>Qdesc</i> | Leaf parent H | 2,976.6      | 21x   | 23x   | 3.5    | 0.07x | 0.13x | 2.3     | 0.89x | 0.90x |
| WT100K | <i>Qdesc</i> | Leaf parent   | 575.1        | 15x   | 300x  | 2.2    | 0.04x | 0.08x | 1.2     | 0.93x | 0.92x |
| SNB/C  | <i>Qdesc</i> | Root          | 91,997.5     | 140x  | 47Kx  | 3.6    | 0.01x | 0.01x | 4.7     | 1.0x  | 1.0x  |
| SNB/C  | <i>Qdesc</i> | Deep root     | 81,661.1     | 31x   | 39x   | 4.1    | 0.01x | 0.01x | 4.1     | 0.92x | 1.0x  |
| SNB/C  | <i>Qdesc</i> | Leaf parent H | 68,727.3     | 26x   | 28x   | 3.6    | 0.01x | 0.01x | 3.5     | 0.93x | 0.87x |
| SNB/C  | <i>Qdesc</i> | Leaf parent   | 22,391.0     | 31x   | 11Kx  | 3.4    | 0.01x | 0.01x | 1.6     | 1.0x  | 0.99x |
| SNB/P  | <i>Qdesc</i> | Root          | 16,838.8     | 6.2Kx | 6.6Kx | 6.9    | 2.8x  | 3.1x  | 46.3    | 1.1x  | 1.1x  |
| SNB/P  | <i>Qdesc</i> | Leaf parent H | 15,130.5     | 5.1Kx | 5.1Kx | 3.7    | 1.9x  | 2.0x  | 10.9    | 0.85x | 0.79x |
| SNB/P  | <i>Qdesc</i> | Leaf parent   | 14,506.6     | 9.0Kx | 7.8Kx | 3.8    | 1.8x  | 2.0x  | 1.1     | 0.87x | 0.74x |
| SNB/T  | <i>Qdesc</i> | Root          | 14,465.3     | 7.7Kx | 7.6Kx | 3.3    | 2.0x  | 2.1x  | 9.7     | 0.89x | 0.85x |
| SNB/T  | <i>Qdesc</i> | Leaf parent H | 14,542.6     | 9.9Kx | 10Kx  | 3.1    | 1.9x  | 1.8x  | 8.6     | 0.96x | 0.94x |
| SNB/T  | <i>Qdesc</i> | Leaf parent   | 14,497.3     | 9.0Kx | 8.6Kx | 2.9    | 1.9x  | 2.0x  | 1.5     | 1.0x  | 1.0x  |

| Graph  | Query                    | Parameters  | Apache AGE |       |       | Kuzu   |       |       | Neo4j  |       |       |
|--------|--------------------------|-------------|------------|-------|-------|--------|-------|-------|--------|-------|-------|
|        |                          |             | B (ms)     | SD    | SP    | B (ms) | SD    | SP    | B (ms) | SD    | SP    |
| F40    | <i>Q<sub>child</sub></i> | High degree | 2.0        | 1.2x  | 1.3x  | 1.7    | 0.91x | 0.93x | 1.7    | 1.1x  | 1.0x  |
| F40    | <i>Q<sub>child</sub></i> | Low degree  | 1.9        | 1.2x  | 1.3x  | 2.0    | 1.2x  | 1.0x  | 1.2    | 1.1x  | 1.1x  |
| F1K    | <i>Q<sub>child</sub></i> | High degree | 2.4        | 0.97x | 1.0x  | 1.6    | 0.63x | 0.64x | 1.9    | 0.99x | 1.2x  |
| F1K    | <i>Q<sub>child</sub></i> | Low degree  | 2.3        | 0.95x | 1.0x  | 1.5    | 0.65x | 1.1x  | 1.1    | 0.97x | 0.96x |
| NT10   | <i>Q<sub>child</sub></i> | High degree | 1.9        | 1.2x  | 1.5x  | 1.7    | 0.82x | 0.89x | 1.6    | 0.94x | 0.94x |
| NT10   | <i>Q<sub>child</sub></i> | Low degree  | 1.9        | 1.2x  | 1.3x  | 1.6    | 0.94x | 1.3x  | 1.3    | 1.2x  | 1.2x  |
| NT100  | <i>Q<sub>child</sub></i> | High degree | 2.0        | 1.3x  | 1.3x  | 1.8    | 1.1x  | 1.1x  | 2.0    | 0.97x | 0.91x |
| NT100  | <i>Q<sub>child</sub></i> | Low degree  | 1.9        | 1.2x  | 1.3x  | 1.7    | 1.0x  | 0.95x | 1.3    | 1.1x  | 0.89x |
| NT1K   | <i>Q<sub>child</sub></i> | High degree | 2.4        | 1.0x  | 1.0x  | 1.7    | 0.89x | 0.88x | 1.2    | 1.1x  | 1.1x  |
| NT1K   | <i>Q<sub>child</sub></i> | Low degree  | 2.2        | 0.97x | 1.0x  | 1.7    | 0.91x | 0.96x | 0.7    | 0.97x | 1.0x  |
| NT10K  | <i>Q<sub>child</sub></i> | High degree | 6.3        | 0.60x | 0.61x | 2.2    | 0.49x | 0.72x | 1.9    | 0.98x | 1.0x  |
| NT10K  | <i>Q<sub>child</sub></i> | Low degree  | 5.4        | 0.52x | 0.53x | 2.2    | 0.43x | 0.91x | 1.1    | 0.94x | 0.97x |
| NT100K | <i>Q<sub>child</sub></i> | High degree | 57.3       | 0.53x | 0.55x | 2.2    | 0.06x | 0.08x | 3.5    | 0.92x | 0.86x |
| NT100K | <i>Q<sub>child</sub></i> | Low degree  | 55.8       | 0.53x | 0.54x | 1.9    | 0.05x | 0.07x | 1.3    | 0.93x | 0.98x |
| DT10   | <i>Q<sub>child</sub></i> | High degree | 2.0        | 1.3x  | 1.3x  | 1.8    | 1.3x  | 1.2x  | 1.5    | 0.89x | 0.95x |
| DT10   | <i>Q<sub>child</sub></i> | Low degree  | 1.9        | 1.2x  | 1.3x  | 1.6    | 0.90x | 0.92x | 1.1    | 0.97x | 0.97x |
| DT100  | <i>Q<sub>child</sub></i> | High degree | 2.0        | 1.2x  | 1.3x  | 1.4    | 0.99x | 0.82x | 1.7    | 1.1x  | 1.1x  |
| DT100  | <i>Q<sub>child</sub></i> | Low degree  | 1.9        | 1.2x  | 1.3x  | 1.9    | 1.2x  | 1.1x  | 1.0    | 0.94x | 0.94x |
| DT1K   | <i>Q<sub>child</sub></i> | High degree | 2.4        | 1.0x  | 1.1x  | 1.1    | 0.64x | 0.60x | 1.5    | 0.98x | 1.1x  |
| DT1K   | <i>Q<sub>child</sub></i> | Low degree  | 2.3        | 0.94x | 1.0x  | 1.8    | 1.0x  | 1.0x  | 1.0    | 0.93x | 0.92x |
| DT10K  | <i>Q<sub>child</sub></i> | High degree | 6.3        | 0.60x | 0.61x | 1.7    | 0.34x | 0.59x | 1.5    | 1.0x  | 1.0x  |
| DT10K  | <i>Q<sub>child</sub></i> | Low degree  | 5.4        | 0.52x | 0.53x | 1.8    | 0.41x | 0.67x | 1.0    | 1.0x  | 1.0x  |
| DT100K | <i>Q<sub>child</sub></i> | High degree | 65.5       | 0.62x | 0.66x | 1.7    | 0.04x | 0.08x | 1.8    | 1.0x  | 1.1x  |
| DT100K | <i>Q<sub>child</sub></i> | Low degree  | 56.3       | 0.53x | 0.54x | 1.8    | 0.05x | 0.09x | 1.2    | 0.90x | 1.0x  |

| Graph  | Query       | Parameters  | Apache AGE |       |       | Kuzu   |       |       | Neo4j  |       |       |
|--------|-------------|-------------|------------|-------|-------|--------|-------|-------|--------|-------|-------|
|        |             |             | B (ms)     | SD    | SP    | B (ms) | SD    | SP    | B (ms) | SD    | SP    |
| WT10   | $Q_{child}$ | High degree | 2.0        | 1.3x  | 1.4x  | 2.2    | 1.2x  | 1.3x  | 1.7    | 0.97x | 0.99x |
| WT10   | $Q_{child}$ | Low degree  | 1.8        | 1.2x  | 1.3x  | 1.7    | 1.0x  | 1.3x  | 1.1    | 0.99x | 0.95x |
| WT100  | $Q_{child}$ | High degree | 2.0        | 1.2x  | 1.3x  | 1.6    | 0.92x | 0.87x | 2.0    | 0.91x | 1.1x  |
| WT100  | $Q_{child}$ | Low degree  | 2.0        | 1.2x  | 1.3x  | 1.6    | 1.1x  | 1.1x  | 1.1    | 0.98x | 0.96x |
| WT1K   | $Q_{child}$ | High degree | 2.5        | 1.0x  | 1.0x  | 1.6    | 0.89x | 0.86x | 2.3    | 0.95x | 0.99x |
| WT1K   | $Q_{child}$ | Low degree  | 2.3        | 1.0x  | 0.99x | 1.6    | 0.93x | 1.0x  | 1.1    | 0.95x | 0.92x |
| WT10K  | $Q_{child}$ | High degree | 6.3        | 0.59x | 0.60x | 1.5    | 0.29x | 0.51x | 2.6    | 0.88x | 0.96x |
| WT10K  | $Q_{child}$ | Low degree  | 5.4        | 0.52x | 0.52x | 1.5    | 0.35x | 0.58x | 1.2    | 0.98x | 0.98x |
| WT100K | $Q_{child}$ | High degree | 63.0       | 0.62x | 0.63x | 1.6    | 0.04x | 0.06x | 1.5    | 0.87x | 0.84x |
| WT100K | $Q_{child}$ | Low degree  | 54.2       | 0.52x | 0.55x | 2.0    | 0.05x | 0.09x | 0.7    | 1.1x  | 1.1x  |
| SNB/C  | $Q_{child}$ | High degree | 1,427.9    | 0.40x | 0.41x | 2.8    | 0.01x | 0.01x | 3.4    | 0.90x | 0.89x |
| SNB/C  | $Q_{child}$ | Low degree  | 1,169.3    | 0.33x | 0.36x | 1.8    | 0.01x | 0.01x | 1.2    | 0.92x | 1.2x  |
| SNB/P  | $Q_{child}$ | High degree | 3.5        | 1.1x  | 1.2x  | 2.4    | 1.0x  | 1.1x  | 24.4   | 0.95x | 0.93x |
| SNB/P  | $Q_{child}$ | Low degree  | 2.4        | 0.82x | 0.94x | 1.5    | 0.62x | 0.63x | 1.2    | 0.91x | 0.92x |
| SNB/T  | $Q_{child}$ | High degree | 2.1        | 1.3x  | 1.4x  | 1.7    | 1.0x  | 0.94x | 3.7    | 0.89x | 0.90x |
| SNB/T  | $Q_{child}$ | Low degree  | 1.9        | 1.2x  | 1.3x  | 1.7    | 1.1x  | 1.0x  | 1.2    | 1.1x  | 0.94x |

| Graph  | Query        | Parameters      | Apache AGE   |                |                | Kuzu   |                |                | Neo4j   |                |                |
|--------|--------------|-----------------|--------------|----------------|----------------|--------|----------------|----------------|---------|----------------|----------------|
|        |              |                 | B (ms)       | S <sub>D</sub> | S <sub>P</sub> | B (ms) | S <sub>D</sub> | S <sub>P</sub> | B (ms)  | S <sub>D</sub> | S <sub>P</sub> |
| F40    | <i>Qleaf</i> | Root            | 9.6          | 6.1x           | 4.7x           | 5.3    | 1.7x           | 3.2x           | 1.9     | 1.0x           | 1.1x           |
| F40    | <i>Qleaf</i> | Leaf parent H   | 9.9          | 6.5x           | 4.8x           | 4.6    | 1.7x           | 2.3x           | 1.4     | 0.94x          | 0.94x          |
| F40    | <i>Qleaf</i> | Leaf parent     | 9.6          | 6.5x           | 4.7x           | 4.2    | 1.3x           | 2.2x           | 1.4     | 0.96x          | 0.84x          |
| F1K    | <i>Qleaf</i> | Root            | 305.9        | 16x            | 130x           | 8.0    | 2.7x           | 3.6x           | 11.3    | 0.43x          | 0.43x          |
| F1K    | <i>Qleaf</i> | Leaf parent H   | 96.2         | 40x            | 43x            | 5.6    | 1.6x           | 2.9x           | 2.3     | 0.86x          | 1.1x           |
| F1K    | <i>Qleaf</i> | Leaf parent     | 95.0         | 40x            | 42x            | 6.4    | 2.4x           | 3.6x           | 1.9     | 0.96x          | 1.1x           |
| NT10   | <i>Qleaf</i> | Root            | 9.7          | 6.6x           | 6.9x           | 3.8    | 1.4x           | 2.0x           | 1.8     | 0.96x          | 0.98x          |
| NT10   | <i>Qleaf</i> | Leaf parent H/L | 9.0          | 6.3x           | 6.3x           | 4.6    | 1.6x           | 2.4x           | 1.6     | 0.93x          | 1.1x           |
| NT100  | <i>Qleaf</i> | Root            | 18.0         | 6.4x           | 2.9x           | 6.1    | 2.0x           | 3.8x           | 9.4     | 0.84x          | 0.96x          |
| NT100  | <i>Qleaf</i> | Leaf parent H   | 11.2         | 7.4x           | 1.8x           | 5.0    | 1.7x           | 2.8x           | 2.2     | 0.91x          | 1.1x           |
| NT100  | <i>Qleaf</i> | Leaf parent     | 10.9         | 7.2x           | 1.8x           | 5.4    | 1.8x           | 3.3x           | 1.8     | 1.0x           | 1.1x           |
| NT1K   | <i>Qleaf</i> | Root            | 946.0        | 7.5x           | 340x           | 12.5   | 3.9x           | 5.6x           | 30.6    | 0.88x          | 0.87x          |
| NT1K   | <i>Qleaf</i> | Leaf parent H   | 161.4        | 71x            | 70x            | 4.5    | 1.2x           | 2.2x           | 1.4     | 0.86x          | 0.96x          |
| NT1K   | <i>Qleaf</i> | Leaf parent     | 158.7        | 65x            | 69x            | 5.2    | 1.5x           | 3.0x           | 1.2     | 1.0x           | 1.1x           |
| NT10K  | <i>Qleaf</i> | Root            | 92,769.6     | 7.4x           | 6.4Kx          | 48.5   | 2.4x           | 6.4x           | 294.7   | 0.87x          | 0.84x          |
| NT10K  | <i>Qleaf</i> | Leaf parent H   | 11,983.1     | 1.0Kx          | 1.1Kx          | 5.7    | 0.32x          | 2.3x           | 1.4     | 0.75x          | 0.66x          |
| NT10K  | <i>Qleaf</i> | Leaf parent     | 11,977.1     | 1.0Kx          | 1.1Kx          | 5.1    | 0.32x          | 2.2x           | 1.8     | 0.92x          | 0.96x          |
| NT100K | <i>Qleaf</i> | Root            | 19,008,309.3 | 16x            | 170Kx          | 108.7  | 0.53x          | 1.5x           | 4,085.6 | 0.83x          | 0.81x          |
| NT100K | <i>Qleaf</i> | Leaf parent H   | 1,230,113.2  | 3.6Kx          | 11Kx           | 5.7    | 0.04x          | 0.27x          | 2.4     | 0.97x          | 1.0x           |
| NT100K | <i>Qleaf</i> | Leaf parent     | 1,227,753.7  | 3.6Kx          | 11Kx           | 4.6    | 0.03x          | 0.23x          | 1.3     | 1.0x           | 1.2x           |
| DT10   | <i>Qleaf</i> | Root            | 9.4          | 6.2x           | 6.5x           | 3.9    | 1.5x           | 2.3x           | 1.8     | 0.97x          | 0.90x          |
| DT10   | <i>Qleaf</i> | Leaf parent H   | 9.6          | 6.8x           | 6.7x           | 3.6    | 1.4x           | 2.2x           | 2.0     | 1.0x           | 1.0x           |
| DT10   | <i>Qleaf</i> | Leaf parent     | 9.7          | 7.4x           | 6.6x           | 3.9    | 1.5x           | 2.3x           | 1.5     | 0.95x          | 1.0x           |
| DT100  | <i>Qleaf</i> | Root            | 16.2         | 5.7x           | 3.0x           | 4.5    | 1.6x           | 2.5x           | 6.8     | 0.75x          | 0.83x          |
| DT100  | <i>Qleaf</i> | Leaf parent H   | 11.1         | 7.3x           | 2.1x           | 4.1    | 1.6x           | 2.5x           | 1.8     | 0.94x          | 1.0x           |
| DT100  | <i>Qleaf</i> | Leaf parent     | 10.8         | 7.1x           | 2.0x           | 4.1    | 1.4x           | 2.3x           | 1.6     | 0.96x          | 1.0x           |
| DT1K   | <i>Qleaf</i> | Root            | 645.4        | 5.4x           | 240x           | 13.1   | 4.7x           | 6.5x           | 37.6    | 1.1x           | 1.3x           |
| DT1K   | <i>Qleaf</i> | Leaf parent H   | 97.0         | 40x            | 43x            | 4.8    | 1.5x           | 2.4x           | 1.3     | 0.72x          | 0.72x          |
| DT1K   | <i>Qleaf</i> | Leaf parent     | 96.3         | 41x            | 42x            | 5.1    | 1.8x           | 2.8x           | 1.6     | 0.99x          | 1.0x           |
| DT10K  | <i>Qleaf</i> | Root            | 59,616.8     | 5.5x           | 4.5Kx          | 30.4   | 2.9x           | 5.7x           | 197.5   | 0.87x          | 0.86x          |
| DT10K  | <i>Qleaf</i> | Leaf parent H   | 6,430.0      | 550x           | 610x           | 3.8    | 0.29x          | 2.1x           | 1.3     | 0.81x          | 0.70x          |
| DT10K  | <i>Qleaf</i> | Leaf parent     | 6,417.6      | 550x           | 610x           | 5.2    | 0.29x          | 2.7x           | 1.6     | 1.0x           | 1.0x           |
| DT100K | <i>Qleaf</i> | Root            | 11,677,527.2 | 13x            | 120Kx          | 135.9  | 0.87x          | 3.2x           | 2,118.7 | 0.83x          | 0.81x          |
| DT100K | <i>Qleaf</i> | Leaf parent H   | 617,168.0    | 1.8Kx          | 5.6Kx          | 5.9    | 0.05x          | 0.51x          | 1.3     | 0.88x          | 0.98x          |
| DT100K | <i>Qleaf</i> | Leaf parent     | 617,092.0    | 1.8Kx          | 5.7Kx          | 4.5    | 0.04x          | 0.36x          | 1.1     | 0.94x          | 1.1x           |

| Graph  | Query        | Parameters    | Apache AGE   |        |        | Kuzu   |       |       | Neo4j   |       |       |
|--------|--------------|---------------|--------------|--------|--------|--------|-------|-------|---------|-------|-------|
|        |              |               | B (ms)       | SD     | SP     | B (ms) | SD    | SP    | B (ms)  | SD    | SP    |
| WT10   | <i>Qleaf</i> | Root          | 9.6          | 6.4x   | 6.5x   | 4.1    | 1.5x  | 2.6x  | 1.9     | 0.96x | 1.0x  |
| WT10   | <i>Qleaf</i> | Leaf parent H | 9.2          | 6.5x   | 6.4x   | 4.3    | 1.5x  | 2.3x  | 1.6     | 0.88x | 0.90x |
| WT10   | <i>Qleaf</i> | Leaf parent   | 9.6          | 6.7x   | 6.7x   | 3.7    | 1.5x  | 2.5x  | 1.5     | 0.94x | 0.99x |
| WT100  | <i>Qleaf</i> | Root          | 20.3         | 7.1x   | 2.7x   | 5.0    | 1.7x  | 2.7x  | 7.3     | 0.65x | 0.61x |
| WT100  | <i>Qleaf</i> | Leaf parent H | 12.0         | 7.6x   | 1.6x   | 3.7    | 1.4x  | 1.9x  | 2.8     | 0.91x | 0.94x |
| WT100  | <i>Qleaf</i> | Leaf parent   | 11.1         | 7.1x   | 1.5x   | 3.8    | 1.4x  | 2.0x  | 2.1     | 0.94x | 0.98x |
| WT1K   | <i>Qleaf</i> | Root          | 1,134.1      | 8.4x   | 410x   | 9.4    | 3.0x  | 4.9x  | 45.1    | 1.1x  | 1.1x  |
| WT1K   | <i>Qleaf</i> | Leaf parent H | 169.6        | 70x    | 73x    | 4.7    | 1.4x  | 2.4x  | 1.7     | 0.94x | 0.98x |
| WT1K   | <i>Qleaf</i> | Leaf parent   | 167.4        | 69x    | 72x    | 4.8    | 1.4x  | 3.0x  | 1.5     | 0.94x | 0.98x |
| WT10K  | <i>Qleaf</i> | Root          | 110,118.3    | 8.1x   | 7.2Kx  | 29.8   | 1.2x  | 3.7x  | 376.1   | 0.94x | 0.82x |
| WT10K  | <i>Qleaf</i> | Leaf parent H | 12,540.9     | 1.1Kx  | 1.1Kx  | 3.7    | 0.18x | 1.5x  | 2.1     | 0.78x | 0.64x |
| WT10K  | <i>Qleaf</i> | Leaf parent   | 12,518.4     | 1.1Kx  | 1.1Kx  | 4.5    | 0.21x | 1.8x  | 2.8     | 1.0x  | 1.3x  |
| WT100K | <i>Qleaf</i> | Root          | 18,195,767.1 | 17x    | 160Kx  | 114.9  | 0.58x | 1.6x  | 3,990.9 | 0.84x | 0.81x |
| WT100K | <i>Qleaf</i> | Leaf parent H | 1,227,203.3  | 3.6Kx  | 12Kx   | 5.4    | 0.03x | 0.21x | 2.3     | 0.78x | 0.54x |
| WT100K | <i>Qleaf</i> | Leaf parent   | 1,224,686.1  | 3.6Kx  | 11Kx   | 4.3    | 0.03x | 0.21x | 1.8     | 0.90x | 0.94x |
| SNB/C  | <i>Qleaf</i> | Root          | >6 h         | >8.0Kx | >9.2Kx | 6.0    | 0.01x | 0.02x | 2.0     | 0.66x | 0.70x |
| SNB/C  | <i>Qleaf</i> | Deep root     | >6 h         | >7.9Kx | >9.2Kx | 5.9    | 0.01x | 0.02x | 2.7     | 0.91x | 0.86x |
| SNB/C  | <i>Qleaf</i> | Leaf parent H | >6 h         | >8.0Kx | >9.2Kx | 5.2    | 0.01x | 0.01x | 3.6     | 0.93x | 0.93x |
| SNB/C  | <i>Qleaf</i> | Leaf parent   | >6 h         | >7.9Kx | >9.2Kx | 5.9    | 0.01x | 0.02x | 1.6     | 0.98x | 1.0x  |
| SNB/P  | <i>Qleaf</i> | Root          | 16,989.0     | 210x   | 5.3Kx  | 9.6    | 2.2x  | 3.8x  | 46.1    | 1.1x  | 1.1x  |
| SNB/P  | <i>Qleaf</i> | Leaf parent H | 15,466.6     | 1.8Kx  | 5.2Kx  | 5.1    | 1.3x  | 2.4x  | 11.7    | 0.81x | 0.76x |
| SNB/P  | <i>Qleaf</i> | Leaf parent   | 14,814.6     | 5.1Kx  | 5.3Kx  | 5.0    | 1.2x  | 2.2x  | 1.2     | 0.80x | 0.80x |
| SNB/T  | <i>Qleaf</i> | Root          | 14,533.6     | 6.4Kx  | 3.3Kx  | 4.6    | 1.9x  | 2.6x  | 10.2    | 1.0x  | 1.1x  |
| SNB/T  | <i>Qleaf</i> | Leaf parent H | 14,546.1     | 7.3Kx  | 3.4Kx  | 4.4    | 1.7x  | 2.6x  | 8.2     | 1.3x  | 0.95x |
| SNB/T  | <i>Qleaf</i> | Leaf parent   | 14,430.3     | 9.5Kx  | 3.4Kx  | 3.4    | 1.4x  | 2.0x  | 1.7     | 1.1x  | 1.1x  |

| Graph  | Query           | Parameters       | Apache AGE |                |                | Kuzu   |                |                | Neo4j  |                |                |
|--------|-----------------|------------------|------------|----------------|----------------|--------|----------------|----------------|--------|----------------|----------------|
|        |                 |                  | B (ms)     | S <sub>D</sub> | S <sub>P</sub> | B (ms) | S <sub>D</sub> | S <sub>P</sub> | B (ms) | S <sub>D</sub> | S <sub>P</sub> |
| F40    | <i>Qa&amp;d</i> | Root-leaf        | 10.1       | 7.3x           | 7.9x           | 22.2   | 25x            | 24x            | 1.5    | 1.0x           | 1.1x           |
| F40    | <i>Qa&amp;d</i> | Deep parent-leaf | 10.1       | 9.2x           | 8.1x           | 19.9   | 22x            | 21x            | 1.3    | 0.98x          | 0.87x          |
| F40    | <i>Qa&amp;d</i> | Siblings         | 9.9        | 9.0x           | 9.1x           | 20.1   | 21x            | 22x            | 1.5    | 1.2x           | 1.1x           |
| F40    | <i>Qa&amp;d</i> | Distant leaves   | 9.3        | 6.7x           | 8.5x           | 21.4   | 24x            | 23x            | 1.5    | 1.1x           | 1.0x           |
| F1K    | <i>Qa&amp;d</i> | Root-leaf        | 12.5       | 8.8x           | 8.4x           | 19.1   | 19x            | 21x            | 1.7    | 1.5x           | 1.1x           |
| F1K    | <i>Qa&amp;d</i> | Deep parent-leaf | 12.2       | 7.5x           | 8.4x           | 21.6   | 22x            | 24x            | 1.8    | 1.6x           | 1.2x           |
| F1K    | <i>Qa&amp;d</i> | Siblings         | 10.8       | 4.2x           | 4.2x           | 16.9   | 16x            | 19x            | 1.6    | 1.2x           | 0.96x          |
| F1K    | <i>Qa&amp;d</i> | Distant leaves   | 10.4       | 4.1x           | 4.0x           | 22.3   | 23x            | 21x            | 1.7    | 1.3x           | 1.0x           |
| NT10   | <i>Qa&amp;d</i> | Root-leaf        | 9.8        | 8.0x           | 9.1x           | 18.7   | 20x            | 20x            | 1.6    | 1.2x           | 1.3x           |
| NT10   | <i>Qa&amp;d</i> | Deep parent-leaf | 9.8        | 8.9x           | 7.9x           | 19.0   | 19x            | 20x            | 1.3    | 1.1x           | 1.2x           |
| NT10   | <i>Qa&amp;d</i> | Siblings         | 9.3        | 7.3x           | 9.5x           | 20.9   | 21x            | 21x            | 1.5    | 1.0x           | 0.87x          |
| NT10   | <i>Qa&amp;d</i> | Distant leaves   | 9.4        | 9.4x           | 9.4x           | 20.4   | 23x            | 22x            | 1.4    | 1.2x           | 1.1x           |
| NT100  | <i>Qa&amp;d</i> | Root-leaf        | 10.0       | 7.6x           | 7.6x           | 20.7   | 22x            | 23x            | 2.0    | 1.5x           | 2.7x           |
| NT100  | <i>Qa&amp;d</i> | Deep parent-leaf | 10.1       | 7.9x           | 7.7x           | 20.7   | 22x            | 21x            | 1.4    | 1.1x           | 0.96x          |
| NT100  | <i>Qa&amp;d</i> | Siblings         | 9.3        | 8.0x           | 8.2x           | 18.7   | 20x            | 20x            | 1.7    | 1.3x           | 1.5x           |
| NT100  | <i>Qa&amp;d</i> | Distant leaves   | 9.5        | 8.3x           | 8.5x           | 21.8   | 24x            | 23x            | 1.2    | 1.3x           | 1.3x           |
| NT1K   | <i>Qa&amp;d</i> | Root-leaf        | 10.8       | 7.4x           | 7.9x           | 19.5   | 20x            | 22x            | 1.2    | 1.5x           | 1.7x           |
| NT1K   | <i>Qa&amp;d</i> | Deep parent-leaf | 10.6       | 7.6x           | 8.1x           | 19.3   | 23x            | 22x            | 0.9    | 0.78x          | 0.79x          |
| NT1K   | <i>Qa&amp;d</i> | Siblings         | 10.8       | 4.2x           | 3.9x           | 22.4   | 24x            | 26x            | 1.4    | 1.3x           | 1.2x           |
| NT1K   | <i>Qa&amp;d</i> | Distant leaves   | 10.6       | 4.2x           | 4.1x           | 20.6   | 23x            | 22x            | 1.4    | 1.4x           | 1.5x           |
| NT10K  | <i>Qa&amp;d</i> | Root-leaf        | 18.2       | 12x            | 13x            | 22.5   | 26x            | 24x            | 1.6    | 1.2x           | 1.2x           |
| NT10K  | <i>Qa&amp;d</i> | Deep parent-leaf | 18.1       | 12x            | 11x            | 22.5   | 26x            | 26x            | 1.3    | 1.2x           | 1.2x           |
| NT10K  | <i>Qa&amp;d</i> | Siblings         | 21.2       | 1.3x           | 1.3x           | 21.0   | 24x            | 23x            | 1.4    | 1.1x           | 1.1x           |
| NT10K  | <i>Qa&amp;d</i> | Distant leaves   | 21.0       | 1.3x           | 1.3x           | 24.7   | 30x            | 26x            | 1.6    | 1.3x           | 1.3x           |
| NT100K | <i>Qa&amp;d</i> | Root-leaf        | 78.5       | 49x            | 48x            | 19.8   | 18x            | 23x            | 1.1    | 1.2x           | 1.4x           |
| NT100K | <i>Qa&amp;d</i> | Deep parent-leaf | 78.7       | 49x            | 49x            | 22.2   | 23x            | 24x            | 0.9    | 1.2x           | 1.1x           |
| NT100K | <i>Qa&amp;d</i> | Siblings         | 125.4      | 0.80x          | 0.80x          | 18.4   | 18x            | 21x            | 0.8    | 1.0x           | 1.0x           |
| NT100K | <i>Qa&amp;d</i> | Distant leaves   | 125.5      | 0.79x          | 0.79x          | 19.7   | 19x            | 21x            | 0.9    | 1.1x           | 1.1x           |
| DT10   | <i>Qa&amp;d</i> | Root-leaf        | 9.8        | 7.3x           | 7.4x           | 17.4   | 20x            | 18x            | 1.4    | 1.2x           | 1.3x           |
| DT10   | <i>Qa&amp;d</i> | Deep parent-leaf | 9.7        | 9.0x           | 7.6x           | 18.4   | 21x            | 18x            | 1.3    | 1.1x           | 1.1x           |
| DT10   | <i>Qa&amp;d</i> | Siblings         | 9.4        | 8.8x           | 9.2x           | 19.1   | 22x            | 20x            | 1.2    | 1.0x           | 1.0x           |
| DT10   | <i>Qa&amp;d</i> | Distant leaves   | 9.4        | 9.2x           | 9.6x           | 18.0   | 21x            | 20x            | 1.3    | 1.1x           | 1.1x           |
| DT100  | <i>Qa&amp;d</i> | Root-leaf        | 9.7        | 8.5x           | 7.3x           | 20.5   | 24x            | 23x            | 1.5    | 1.3x           | 1.4x           |
| DT100  | <i>Qa&amp;d</i> | Deep parent-leaf | 10.2       | 7.5x           | 7.6x           | 21.8   | 23x            | 23x            | 1.2    | 1.1x           | 1.0x           |
| DT100  | <i>Qa&amp;d</i> | Siblings         | 9.5        | 8.0x           | 8.3x           | 19.4   | 19x            | 22x            | 1.3    | 1.1x           | 1.0x           |
| DT100  | <i>Qa&amp;d</i> | Distant leaves   | 9.6        | 8.1x           | 8.5x           | 18.8   | 20x            | 20x            | 1.4    | 1.1x           | 2.1x           |
| DT1K   | <i>Qa&amp;d</i> | Root-leaf        | 11.0       | 7.6x           | 8.6x           | 20.3   | 24x            | 21x            | 1.8    | 1.3x           | 1.5x           |
| DT1K   | <i>Qa&amp;d</i> | Deep parent-leaf | 10.8       | 8.7x           | 9.0x           | 19.9   | 21x            | 21x            | 1.3    | 0.97x          | 1.1x           |
| DT1K   | <i>Qa&amp;d</i> | Siblings         | 11.2       | 4.2x           | 4.4x           | 17.5   | 19x            | 19x            | 1.4    | 1.1x           | 1.0x           |
| DT1K   | <i>Qa&amp;d</i> | Distant leaves   | 10.7       | 4.1x           | 4.2x           | 20.1   | 21x            | 21x            | 1.6    | 1.2x           | 1.2x           |
| DT10K  | <i>Qa&amp;d</i> | Root-leaf        | 18.7       | 12x            | 12x            | 21.9   | 24x            | 21x            | 1.6    | 1.4x           | 1.5x           |
| DT10K  | <i>Qa&amp;d</i> | Deep parent-leaf | 18.0       | 12x            | 11x            | 20.1   | 21x            | 22x            | 1.3    | 1.2x           | 1.2x           |
| DT10K  | <i>Qa&amp;d</i> | Siblings         | 21.5       | 1.3x           | 1.3x           | 19.8   | 23x            | 20x            | 1.4    | 1.1x           | 1.1x           |
| DT10K  | <i>Qa&amp;d</i> | Distant leaves   | 21.2       | 1.3x           | 1.3x           | 20.3   | 21x            | 21x            | 1.6    | 1.3x           | 1.3x           |
| DT100K | <i>Qa&amp;d</i> | Root-leaf        | 80.0       | 49x            | 50x            | 19.7   | 19x            | 19x            | 1.1    | 1.1x           | 1.3x           |
| DT100K | <i>Qa&amp;d</i> | Deep parent-leaf | 80.8       | 60x            | 60x            | 21.3   | 21x            | 23x            | 0.9    | 1.0x           | 0.75x          |
| DT100K | <i>Qa&amp;d</i> | Siblings         | 130.8      | 0.82x          | 0.84x          | 20.0   | 21x            | 21x            | 1.2    | 0.94x          | 0.92x          |
| DT100K | <i>Qa&amp;d</i> | Distant leaves   | 130.2      | 0.80x          | 0.83x          | 21.1   | 25x            | 23x            | 1.4    | 1.1x           | 1.1x           |

| Graph  | Query           | Parameters       | Apache AGE |       |       | Kuzu   |     |     | Neo4j  |       |       |
|--------|-----------------|------------------|------------|-------|-------|--------|-----|-----|--------|-------|-------|
|        |                 |                  | B (ms)     | SD    | SP    | B (ms) | SD  | SP  | B (ms) | SD    | SP    |
| WT10   | <i>Qa&amp;d</i> | Root-leaf        | 10.0       | 7.5x  | 7.8x  | 18.4   | 20x | 21x | 1.8    | 1.6x  | 1.6x  |
| WT10   | <i>Qa&amp;d</i> | Deep parent-leaf | 10.0       | 7.5x  | 7.6x  | 18.9   | 22x | 20x | 1.3    | 1.1x  | 1.1x  |
| WT10   | <i>Qa&amp;d</i> | Siblings         | 9.4        | 9.1x  | 9.6x  | 18.5   | 21x | 20x | 1.4    | 1.0x  | 1.0x  |
| WT10   | <i>Qa&amp;d</i> | Distant leaves   | 9.7        | 9.8x  | 10x   | 17.6   | 19x | 19x | 1.5    | 1.1x  | 1.2x  |
| WT100  | <i>Qa&amp;d</i> | Root-leaf        | 11.4       | 6.7x  | 7.1x  | 19.1   | 21x | 20x | 1.4    | 1.4x  | 1.3x  |
| WT100  | <i>Qa&amp;d</i> | Deep parent-leaf | 11.1       | 7.5x  | 7.6x  | 18.3   | 20x | 19x | 1.2    | 1.1x  | 0.98x |
| WT100  | <i>Qa&amp;d</i> | Siblings         | 9.8        | 8.3x  | 7.7x  | 17.6   | 19x | 19x | 1.3    | 1.1x  | 1.2x  |
| WT100  | <i>Qa&amp;d</i> | Distant leaves   | 9.8        | 8.6x  | 8.8x  | 21.2   | 23x | 22x | 1.7    | 1.2x  | 1.2x  |
| WT1K   | <i>Qa&amp;d</i> | Root-leaf        | 12.7       | 8.1x  | 8.3x  | 20.7   | 25x | 23x | 1.1    | 1.3x  | 0.94x |
| WT1K   | <i>Qa&amp;d</i> | Deep parent-leaf | 12.1       | 7.8x  | 8.0x  | 18.6   | 22x | 21x | 1.3    | 1.0x  | 0.99x |
| WT1K   | <i>Qa&amp;d</i> | Siblings         | 10.8       | 4.2x  | 4.3x  | 21.0   | 24x | 19x | 1.4    | 1.1x  | 1.0x  |
| WT1K   | <i>Qa&amp;d</i> | Distant leaves   | 10.6       | 4.1x  | 4.2x  | 22.6   | 24x | 25x | 1.6    | 1.1x  | 1.2x  |
| WT10K  | <i>Qa&amp;d</i> | Root-leaf        | 19.1       | 11x   | 11x   | 21.2   | 25x | 25x | 1.4    | 1.2x  | 1.5x  |
| WT10K  | <i>Qa&amp;d</i> | Deep parent-leaf | 18.5       | 11x   | 11x   | 21.0   | 22x | 23x | 1.2    | 1.1x  | 1.1x  |
| WT10K  | <i>Qa&amp;d</i> | Siblings         | 20.5       | 1.2x  | 1.3x  | 19.3   | 22x | 23x | 1.3    | 1.1x  | 1.1x  |
| WT10K  | <i>Qa&amp;d</i> | Distant leaves   | 20.9       | 1.3x  | 1.3x  | 20.8   | 25x | 25x | 1.4    | 1.2x  | 1.1x  |
| WT100K | <i>Qa&amp;d</i> | Root-leaf        | 79.8       | 42x   | 44x   | 20.2   | 22x | 23x | 1.6    | 1.2x  | 1.2x  |
| WT100K | <i>Qa&amp;d</i> | Deep parent-leaf | 81.2       | 44x   | 45x   | 22.5   | 27x | 25x | 1.3    | 1.0x  | 0.96x |
| WT100K | <i>Qa&amp;d</i> | Siblings         | 126.2      | 0.78x | 0.80x | 20.0   | 23x | 22x | 1.2    | 0.96x | 0.89x |
| WT100K | <i>Qa&amp;d</i> | Distant leaves   | 125.9      | 0.79x | 0.79x | 19.6   | 22x | 21x | 1.4    | 1.0x  | 0.98x |
| SNB/C  | <i>Qa&amp;d</i> | Root-leaf        | 14,648.6   | 7.3Kx | 7.9Kx | 22.3   | 22x | 24x | 1.6    | 1.3x  | 1.4x  |
| SNB/C  | <i>Qa&amp;d</i> | Deep parent-leaf | 14,582.5   | 6.9Kx | 8.0Kx | 19.4   | 15x | 19x | 1.2    | 1.1x  | 1.0x  |
| SNB/C  | <i>Qa&amp;d</i> | Siblings         | 15,882.2   | 4.4x  | 4.5x  | 18.7   | 18x | 22x | 1.2    | 1.0x  | 1.0x  |
| SNB/C  | <i>Qa&amp;d</i> | Distant leaves   | 15,854.3   | 4.2x  | 4.3x  | 20.3   | 19x | 24x | 1.2    | 1.1x  | 1.1x  |
| SNB/P  | <i>Qa&amp;d</i> | Root-leaf        | 14,525.6   | 13Kx  | 9.7Kx | 19.6   | 24x | 22x | 1.4    | 1.3x  | 1.3x  |
| SNB/P  | <i>Qa&amp;d</i> | Deep parent-leaf | 14,422.9   | 9.2Kx | 9.7Kx | 17.3   | 20x | 19x | 1.3    | 1.2x  | 1.1x  |
| SNB/P  | <i>Qa&amp;d</i> | Siblings         | 14,476.3   | 4.5Kx | 4.6Kx | 20.9   | 24x | 23x | 1.2    | 1.1x  | 1.0x  |
| SNB/P  | <i>Qa&amp;d</i> | Distant leaves   | 14,341.4   | 4.5Kx | 4.6Kx | 17.6   | 20x | 20x | 1.3    | 1.1x  | 1.1x  |
| SNB/T  | <i>Qa&amp;d</i> | Root-leaf        | 14,538.5   | 9.0Kx | 8.6Kx | 18.9   | 23x | 20x | 1.2    | 1.1x  | 1.0x  |
| SNB/T  | <i>Qa&amp;d</i> | Deep parent-leaf | 14,437.5   | 10Kx  | 8.6Kx | 20.5   | 25x | 23x | 1.4    | 1.3x  | 1.3x  |
| SNB/T  | <i>Qa&amp;d</i> | Siblings         | 14,251.4   | 13Kx  | 14Kx  | 19.3   | 23x | 22x | 1.2    | 1.1x  | 1.0x  |
| SNB/T  | <i>Qa&amp;d</i> | Distant leaves   | 14,278.8   | 14Kx  | 14Kx  | 18.3   | 22x | 22x | 1.3    | 0.87x | 1.1x  |

| Graph | Query | Parameters | Apache AGE |                |                | Kuzu   |                |                | Neo4j  |                |                |
|-------|-------|------------|------------|----------------|----------------|--------|----------------|----------------|--------|----------------|----------------|
|       |       |            | B (ms)     | S <sub>D</sub> | S <sub>P</sub> | B (ms) | S <sub>D</sub> | S <sub>P</sub> | B (ms) | S <sub>D</sub> | S <sub>P</sub> |
| SNB   | IC12  | –          | 188,529.8  | 1.1x           | 1.1x           | 396.5  | 1.3x           | 0.98x          | 157.5  | 1.3x           | 1.5x           |
| SNB   | IS2   | –          | 61,383.9   | 16x            | 4.7x           | 43.7   | 0.18x          | 0.16x          | 4.5    | 0.99x          | 0.42x          |
| SNB   | IS6   | –          | 17,296.9   | 8.0x           | 5.3x           | 28.8   | 1.3x           | 0.24x          | 1.9    | 0.95x          | 1.0x           |